

MODEL QUESTION PAPER

TED (10) 5076
(REVISION 2010)

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
OPTICAL INSTRUMENTATION
(EI&IT)

TIME: 3hrs

(MAXIMUM MARKS 100)

PART – A

MARKS

I. Answer the following questions in one or two sentences. Each question carries 2 marks.

- 1.State the laws of reflection.
- 2.Define Total Internal Reflection.
- 3.Define population Inversion.
- 4.Define Hologram.
- 5.Distinguish between Photo Diode and Photo Transistor. (5x2=10)

PART – B

II. Answer any five of the following. Each question carries 6 marks.

- 1.State the conditions for Permanent Interference.
- 2.Distinguish between Fresnel's and Fraunhofer diffraction.
- 3.Explain the construction and operation of Ruby laser.
- 4.Describe the principle of Holography.
- 5.Describe the application of laser in machining process.
- 6.Explain the principle of operation of LED.
- 7.Describe the working of fiber optic displacement sensor. (5x6=30)

PART – C

(Answer one full question from each unit. Each question carries 15 marks)

Unit – I

- III. a. Explain the phenomenon of dispersion in prisms. (7)
b. Explain how Newton's rings are formed. (8)

OR

- IV.a. Explain how two prisms can be combined to have deviation without dispersion. (9)

b. Define the following:

- a) Interference. b) Path and phase difference. c) Diffraction. (6)

Unit – II

V.a. List the characteristics of laser beam. (6)

b. Describe the classification of laser. (9)

OR

VI.a. Explain the working of Nd-YAG laser. (7)

b. Explain the construction and operation of Semiconductor laser. (8)

Unit – III

VII.a. Explain the medical applications of laser. (7)

b. Explain the working of Laser Doppler Velocity meter. (8)

OR

VIII.a. Describe the reconstruction of recorded wave front from the Hologram. (7)

b. Explain the working of Raman Spectroscope. (8)

Unit – IV

IX.a. Explain the working of PIN Diode. (7)

b. Explain the operation of optical data recording. (8)

OR

X.a. Explain the working of fibre optic pressure transducer. (7)

b. Explain the working of laser printer. (8)